From Contextual Expectations to Exclusion

Results draft organized around four empirical claims

October 27, 2026

5 Results

Predicting what a speaker could have said provides a way to generate alternatives. The further question is what listeners do with those alternatives after hearing the speaker's actual utterance. We examine this distinction in four steps: whether an exclusion rule improves on using continuation probabilities directly; why identifying plausible alternatives still leaves exclusion to be explained; which relations between alternatives help in different contexts; and how the focus results compare with scalar inference. The examples below explain the model comparisons, with full numerical results in Appendix A.

5.1 Linking contextual expectations improves exclusion predictions

In the focus experiment, a thirsty listener hears "I only have milk" and considers whether water or juice is available. A next-word model can identify water and juice as plausible alternatives in this context. Using their continuation probabilities directly, however, predicts that they will rarely be excluded. This is the assumption made by No linking: the probability of a candidate completing the neutral response is also its predicted probability of exclusion.

That direct mapping systematically underpredicts human exclusion. Figure 2 shows the mismatch across all 480 trigger–query pairs. Many alternatives receive low continuation probabilities even when participants frequently exclude them. Applying an explicit linking rule changes this pattern. All four rules in the Top-K comparison improve summed log score over No linking; the figure illustrates Disjunction Top-K, which has the highest observed focus score. Its predictions cover more of the range of human judgments, although substantial errors remain. The common benefit is that an alternative's expectedness can contribute to exclusion through a relation among alternatives, rather than serving directly as its exclusion probability.

The improvement is conditional on the mapping. Set Top-p and Conjunction Top-p perform worse than No linking, so introducing structure alone does not guarantee better predictions. Nor do the gains establish strong absolute fit: a reference that predicts the training-fold mean exclusion rate scores above every evaluated focus model. The comparison instead identifies the limitation of using untransformed continuation probabilities and motivates asking what an exclusion rule adds.

5.2 Knowing which alternatives are likely does not determine which are excluded

The BAG context makes this limitation concrete. Someone needs something to write with, and a friend looks in a handbag. The neutral language-model probabilities rank pen above crayons and crayons above chalk. Mean human exclusion rates follow this order as well: the more expected writing implements tend to be more frequently excluded. Yet after "I only have cash," participants excluded all three in every observed response (Figure 3). The model's continuation probabilities are very different from those judgments: pen receives about 1.9%, while crayons and chalk each receive less than .02%.

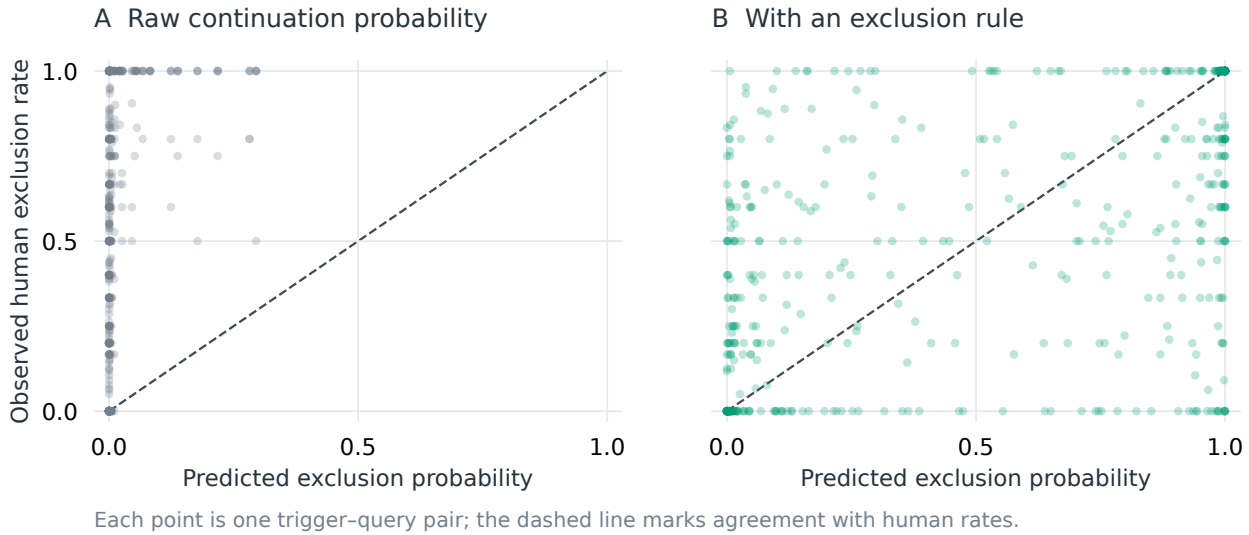


Figure 2: Raw expectedness versus exclusion predictions after linking. Both panels show the same 480 focus trigger–query pairs and use identical probability axes. Panel A uses No linking; Panel B illustrates Disjunction Top-K, the evaluated focus model with the highest aggregate log score. The diagonal marks agreement with observed human rates. All items are retained; points can overlap. The linked model still makes substantial errors, and the comparison is descriptive.



Figure 3: An informative next-word ordering can coexist with poor exclusion probabilities. In BAG, the queried writing implements have different neutral continuation probabilities, yet participants excluded each after the trigger cash in all observed responses. Counts above the human points report the small item-level denominators. These selected cases illustrate the distinction between expected continuations and exclusion judgments; they are not independent estimates of a population effect.

The model can therefore identify useful alternatives without assigning useful probabilities to their exclusion. Continuation probabilities distribute expectation across many possible answers; an utterance with *only* can exclude several of those answers together. Knowing that pen is a more likely continuation than crayons does not mean that crayons should be excluded only rarely after the speaker names cash. This is why positive ranking agreement can coexist with poor probability predictions. Across contexts, No linking has the highest mean exclusion-ranking Spearman among the evaluated focus approaches, even though the four explicit rules in the Top-K comparison have better log scores (Table A1).

The FRIDGE example shows how a linking rule can address the difference. Participants excluded juice after "I only have water" in 22/22 responses and water after "I only have juice" in 6/6. No linking assigns probabilities of about .01 and .06. Set Top-K instead assigns .762 and 1.000, allowing both alternatives to be frequently excluded. It preserves the same exclusion ranking as No linking across the context's 30 pairs while assigning higher probabilities to these judgments. The distinction is between identifying the relative expectedness of alternatives and determining what follows from the speaker's choice among them.

This problem persists when the language model correctly orders the tested words. In BAKERY and FRIDGE, its six-word rankings agree perfectly with the independent human norms. Applying a linking rule still improves log score on those contexts, and the same pattern holds under the broader near-perfect-ranking criterion. These subsets retain the original predictions and boundaries. Agreement about the tested alternatives consequently does not remove the need to explain their exclusion; it also does not imply that the full continuation distribution is correct.

5.3 Membership and relative ordering capture different patterns

The next question is which relation should connect expectations to exclusion. Set allows an alternative to be excluded because it belongs to a contextually available group. Ordering instead asks whether it would precede the uttered trigger in a sampled ordering. The human experiment gives reasons to consider both: mean exclusion is .624 in the diagnostic cases supported by Set and Disjunction, and .414 in those supported by Ordering and Disjunction (Table A5). Comparing the probabilistic models shows where the two relations help and where they mislead.

A case favoring membership: TRANSPORT. Two friends want to go into town without walking. After an utterance naming only a car, participants excluded a motorcycle in 19/21 responses. Set Top-K predicts .830, whereas Ordering predicts .244 (Figure 4A). A motorcycle can belong to the relevant group of transport options even though it is less likely than a car to precede the speaker's choice. Requiring that relative position makes Ordering underpredict its exclusion.

Ordering can also predict exclusion where listeners generally do not. After an utterance naming only chairs, a lawnmower precedes that very unlikely transport answer in almost every sampled ordering. Ordering consequently predicts almost certain exclusion of the lawnmower, while participants excluded it in only 2/22 responses. Set assigns it little membership probability and stays close to the observed rate. Being more expected than an implausible trigger is therefore insufficient to make a query a relevant excluded option. Set has both the better log score and the higher exclusion-ranking correlation across this context's complete 30 pairs.

A case favoring relative ordering: MALL. The pattern differs when two friends are choosing somewhere to have lunch in a shopping mall. After an utterance naming only a burger joint, participants excluded a coffee shop in 10/16 responses. Set Top-K predicts only .038 because the coffee shop rarely enters its sampled sets. Ordering predicts .566: the coffee shop often precedes the burger joint even though its absolute membership probability is low. Here the relative comparison supplies information that a fixed membership boundary misses. Ordering has both the better log score and the higher ranking correlation across all 30 MALL pairs.

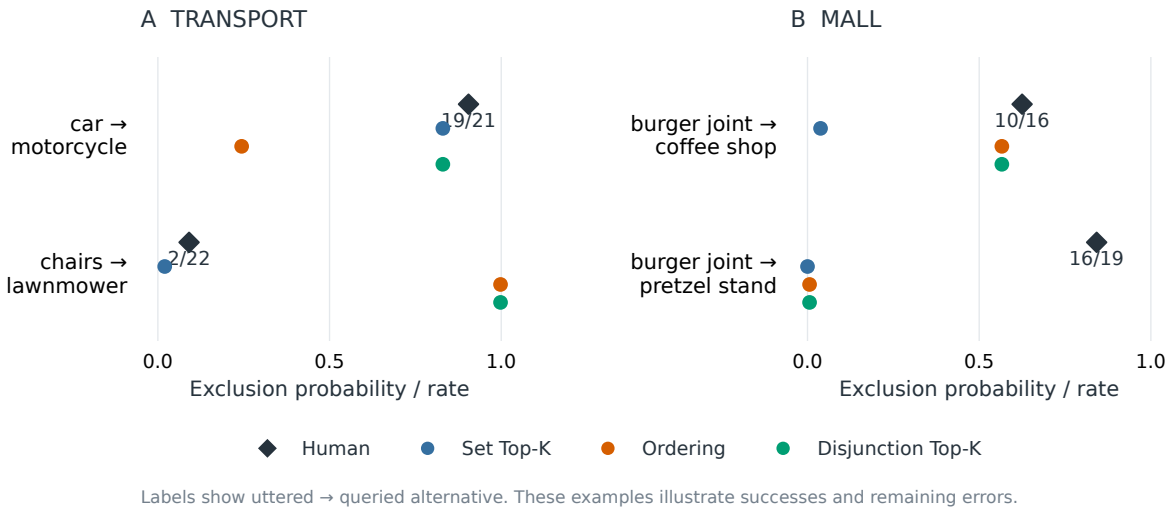


Figure 4: The usefulness of membership and ordering varies across items and contexts. Labels show trigger → query; counts report human exclusions/responses. In TRANSPORT, Set is closer for both motorcycle after car and lawnmower after chairs. In MALL, Ordering is closer for coffee shop after burger joint, while all displayed models severely underpredict exclusion of pretzel stand after that trigger. Disjunction follows Set on the first TRANSPORT item and Ordering on the lawnmower and coffee-shop items, illustrating both a benefit and a cost of allowing either relation to license exclusion. The plotted cases are descriptive examples; full-context comparisons use all 30 pairs (Table A4).

Neither example makes a structure sufficient. In MALL, participants also excluded a pretzel stand after the burger joint in 16/19 responses, but both Set and Ordering assign very low probabilities (Figure 4B). The alternatives supplied by the expectation model can still be inadequate for the exclusion task. Across the 16 focus contexts, Ordering has a higher log score than Set Top-K in twelve, while Set has the higher score in CUT, HOT, MEAT, and TRANSPORT. These are descriptive comparisons of fixed predictions, not evidence that listeners switch rules between contexts.

The combined structures clarify the tradeoff. Conjunction requires both membership and precedence, so either requirement can block an otherwise supported exclusion. Disjunction allows either relation to suffice. It recovers the motorcycle prediction through membership in TRANSPORT and the coffee-shop prediction through ordering in MALL. This flexibility gives Disjunction Top-K the highest aggregate focus log score and the highest score among the four Top-K rules in eleven contexts. It also inherits Ordering's lawnmower error in TRANSPORT. Its aggregate advantage is therefore a balance of gains and errors, rather than a rule that succeeds in every context. The selected Top-p boundaries further expose this dependence on representation: requiring membership performs poorly there, while allowing ordering to provide another route is more robust.

5.4 Contrastive conditioning helps scalar benchmarks but not the focus experiment

The focus task asks which of several contextually relevant options an utterance rules out. The Hu et al. scalar benchmarks instead test conventional weaker–stronger pairs: whether a speaker's use of the weaker expression leads listeners to reject the stronger one. This difference motivates comparing explicit rules applied to neutral expectations with X but not Y, which places the trigger and an explicit contrast inside the language-model prompt.

The contrastive frame has different consequences across the two kinds of data. In each of the four Hu

benchmarks, X but not Y improves log score over No linking. In the focus experiment, it makes the aggregate probability predictions worse (Figure 5). Supplying the trigger and "but not" is thus useful for these conventional scalar comparisons, but does not provide the strongest way to predict exclusion among the contextually generated focus alternatives. In focus, the better predictions come from applying explicit linking rules to neutral expectations.

This contrast is more consistent than the identity of a winning model. Among the Hu datasets, X but not Y leads on log score in Gotzner et al., Set Top-p in van Tiel et al., Ordering in Pankratz and van Tiel, and Disjunction Top-K in Ronai and Xiang (2022). Ranking agreement does not improve uniformly with contrastive conditioning either (Tables A2–A3). The supported distinction concerns how the evaluated expectations are turned into exclusion predictions, not a demonstration that scalar and focus inferences use different human mechanisms.

The Ronai and Xiang (2024) manipulations also show why predicting alternatives and predicting exclusion must remain separate. X but not Y improves log score over No linking in all five conditions and leads in the weak- and strong-QUD conditions. Nevertheless, every evaluated model gives identical predictions when *only* is added to matched items, while human exclusion increases by roughly .30 without an explicit QUD and .27 with a strong QUD. The implementation omits *only* from its scoring prompts and keeps the mapping fixed. Its within-condition performance therefore does not account for the exclusion contributed by the utterance itself. Contextual prediction supplies useful alternatives, but a successful account must also specify how the speaker's expression makes those alternatives excludable.

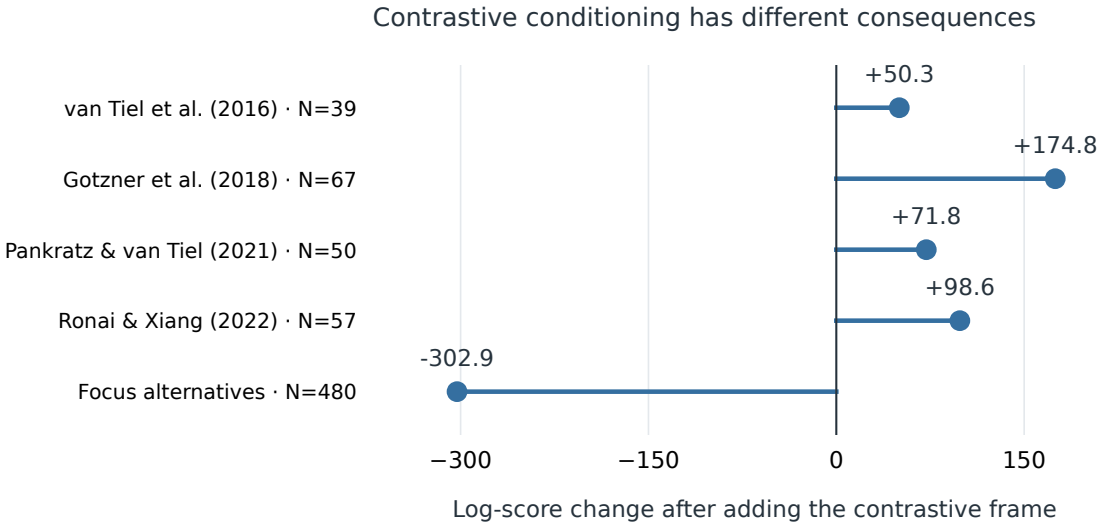


Figure 5: Adding the contrastive frame improves scalar log scores but worsens aggregate focus log score. Values are the summed proper log score of X but not Y minus that of No linking on matching items in each dataset. Positive values favor the contrastive frame. Item counts differ, so magnitudes should not be interpreted as directly comparable effect sizes across datasets. Points and lines show descriptive differences, not uncertainty intervals.

A Supporting result tables

These tables retain the numerical comparisons behind the main argument. All model differences are descriptive.

Approach	Exclusion ρ	Defined contexts	Summed log score
No linking	0.584	16/16	−1544.388
X but not Y	0.521	16/16	−1847.238
Ordering	0.513	16/16	−779.714
Set Top-K	0.567	16/16	−1102.600
Conjunction Top-K	0.569	16/16	−1170.137
Disjunction Top-K	0.528	16/16	−729.351
Set Top-p	0.553	15/16	−1633.002
Conjunction Top-p	0.565	15/16	−1694.071
Disjunction Top-p	0.531	16/16	−732.908

Table A1: Focus model comparisons on 480 pairs. Spearman values are means of defined within-context correlations over 30 pairs; log scores sum all 480 item contributions. Top-p Set and Conjunction have undefined correlation in one context, so their rank means cover 15 contexts. Ordering has no set boundary. The training-fold constant-rate reference has summed log score −334.000.

Dataset / condition	No linking	X but not Y	Set Top-K	Set Top-p	Ordering	Conj. Top-K	Conj. Top-p	Disj. Top-K	Disj. Top-p
van Tiel (2016)	−113.229	−62.936	−70.858	−48.023	−48.979	−85.616	−63.610	−48.423	−50.283
Gotzner (2018)	−309.738	−134.909	−394.037	−458.384	−190.374	−394.303	−458.404	−190.235	−190.466
Pankratz & van Tiel (2021)	−137.042	−65.196	−166.461	−139.888	−61.911	−163.818	−108.959	−66.037	−96.007
R&X (2022)	−174.336	−75.688	−142.247	−175.024	−63.311	−143.390	−175.186	−63.059	−65.223
R&X Baseline	−183.392	−73.146	−162.730	−198.655	−68.489	−150.801	−184.629	−81.356	−84.740
R&X Weak QUD	−107.097	−79.663	−125.890	−198.448	−102.122	−102.399	−105.752	−126.142	−194.895
R&X Strong QUD	−81.824	−56.702	−307.326	−456.516	−86.907	−77.445	−86.595	−357.789	−461.567
R&X Only	−333.149	−153.979	−315.970	−307.110	−95.594	−323.693	−314.892	−88.648	−89.906
R&X Strong QUD + only	−107.959	−49.439	−96.480	−137.559	−39.065	−37.425	−43.521	−109.177	−133.711

Table A2: Summed proper log scores within each scalar dataset or condition. Compare models on the matching items within each row; totals depend on item count. Hu datasets contain 39, 67, 50, and 57 scales, respectively. Each Ronai and Xiang (2024) condition contains the same 60 scales.

Dataset / condition	No linking	X but not Y	Set Top-K	Set Top-p	Ordering	Conj. Top-K	Conj. Top-p	Disj. Top-K	Disj. Top-p
van Tiel (2016)	0.100	0.139	0.166	0.067	−0.040	0.148	0.020	−0.032	−0.006
Gotzner (2018)	0.081	0.043	0.092	−0.080	0.076	0.091	−0.078	0.078	0.075
Pankratz & van Tiel (2021)	−0.004	0.216	−0.033	0.003	−0.118	−0.034	−0.002	−0.116	−0.132
R&X (2022)	0.136	0.371	0.221	0.104	0.153	0.247	0.091	0.158	0.202
R&X Baseline	0.163	0.497	0.245	0.107	0.175	0.271	0.111	0.177	0.219
R&X Weak QUD	−0.130	0.335	−0.069	−0.209	−0.083	−0.067	−0.132	−0.074	−0.171
R&X Strong QUD	−0.259	0.032	−0.150	−0.372	0.113	0.094	−0.050	−0.134	−0.350
R&X Only	0.116	0.413	0.221	0.029	0.156	0.215	0.021	0.179	0.207
R&X Strong QUD + only	−0.223	0.060	−0.147	−0.277	−0.036	−0.104	−0.228	−0.139	−0.207

Table A3: Spearman correlations across scales within each scalar dataset or condition. These are distinct from the within-context focus means in Table A1.

Context	Set log score	Ordering log score	Set ρ	Ordering ρ
BAG	−46.725	−25.508	0.860	0.705
BAKERY	−37.954	−32.587	0.912	0.721
BEACH	−42.698	−35.424	0.508	0.498
COLD	−111.849	−75.650	0.199	0.160
CUT	−58.900	−83.993	0.727	0.529
FRIDGE	−30.214	−25.196	0.860	0.769
GYM	−53.429	−37.719	0.538	0.546
HOT	−30.877	−59.375	0.324	0.239
MALL	−182.928	−49.471	0.351	0.555
MASK	−92.688	−73.809	0.323	0.116
MEAT	−43.155	−48.870	-0.031	0.185
RESTAURANT	−81.693	−17.974	0.780	0.768
SALAD	−100.804	−56.841	0.646	0.496
SCIENCE	−63.054	−45.409	0.494	0.616
THROW	−69.464	−33.761	0.855	0.720
TRANSPORT	−56.170	−78.127	0.720	0.586

Table A4: All 16 focus contexts for the Set Top-K/Ordering comparison. Each log score sums 30 item contributions; each Spearman coefficient compares the same 30 pairs. The illustrative TRANSPORT and MALL cases do not define separate fitted models.

Structures predicting exclusion	Human exclusion	Pairs	Responses
All four	0.787	192	852
Set, Disjunction	0.624	48	852
Ordering, Disjunction	0.414	48	852
None	0.213	192	852

Table A5: Human exclusion grouped by categorical structure predictions on the independent norming. All four means Set, Ordering, Conjunction, and Disjunction; None means none predicts exclusion. These groups are not the fitted probabilistic predictions. Means give equal weight to item rates within each condition/context, then to the 16 contexts. Each condition includes 852 responses from 213 participants. The four rows span all 480 pairs and 3,408 judgments.